Zomato Bangalore Restaurant Market Analysis

Data Analytics Portfolio Project

Tools Used: Python, MySQL, Power BI

1. Project Overview

This project analyzes restaurant market data from Bangalore to understand restaurant distribution, customer ratings, dining costs, online ordering availability, and table booking services.

The main purpose of this analysis is to identify restaurant market trends and explore opportunities that may help restaurant owners and food-service businesses make better decisions.

I used Python for data cleaning, MySQL for business analysis, and Power BI to create an interactive dashboard.

2. Business Problem

Bangalore has a competitive restaurant market with different types of restaurants, pricing options, and customer preferences.

Understanding this market can help answer questions such as:

- Which locations have the highest restaurant concentration?
- Which restaurant types are most common?
- How do customer ratings differ across locations and restaurant types?
- How widely are online ordering and table booking services available?
- How do dining costs vary across the market?
- Which locations may offer opportunities for affordable, highly rated restaurants?

3. Project Objectives

The objectives of this project were to:

1. Analyze the distribution of restaurant records across Bangalore.
2. Compare restaurant ratings and average dining costs.
3. Examine online ordering and table booking availability.
4. Understand restaurant distribution across different price segments.
5. Identify potential market opportunities through location-wise comparisons.
6. Present the findings using an interactive Power BI dashboard.

4. Dataset Description

The final cleaned dataset contains **51,717 records** and **11 columns**.

The dataset includes the following information:

Column	Description
cleaned_name	Restaurant name
cleaned_location	Restaurant location
cleaned_ratings	Customer rating
votes	Number of rating votes
cleaned_cost	Approximate dining cost for two
cleaned_cuisines	Cuisine categories
cleaned_rest_type	Restaurant type
online_order	Online ordering availability
book_table	Table booking availability
listed_in(type)	Listing category
listed_in(city)	Listing city

Note: The number of records does not necessarily represent the number of unique restaurants.

5. Data Cleaning Using Python

I used Python to prepare the dataset for analysis.

The cleaning process included:

- Preparing consistent column names for analysis.
- Converting ratings and dining costs into numeric formats.
- Checking missing values.
- Preparing categorical columns for comparison.
- Exporting the cleaned dataset as a CSV file.

During data validation, I identified:

- **10,052 missing rating values**
- **346 missing dining cost values**

I retained these records rather than deleting them because they could still provide useful information for restaurant counts, locations, and service availability.

Missing numeric values were kept as NULL so that averages could be calculated using available data.

6. SQL Analysis Using MySQL

After cleaning the dataset, I imported it into MySQL and wrote **15 business analysis queries**.

The queries focused on:

- Total restaurant records
- Average customer rating
- Average dining cost
- Restaurant distribution by location
- Most common restaurant types
- Online ordering availability
- Table booking availability
- Ratings by online ordering status
- Ratings by table booking status
- Dining cost by restaurant type
- Dining cost by location
- Restaurants with high customer votes
- Restaurant distribution by price segment
- Locations with affordable dining options
- Locations with strong ratings and reasonable costs

This stage helped me practice SQL aggregation, filtering, grouping, sorting, and business-oriented analysis.

7. Power BI Dashboard

I created an interactive Power BI report with three pages.

Page 1: Market Overview

This page provides a general overview of Bangalore's restaurant market.

Key elements:

- Total Restaurant Records
- Average Customer Rating
- Average Dining Cost
- Online Ordering Percentage
- Table Booking Percentage
- Top 10 Locations by Restaurant Count
- Top 5 Restaurant Types
- Interactive location and restaurant-type filters

Purpose: To understand the overall size and structure of the restaurant market.

Page 2: Restaurant Performance

This page focuses on customer ratings and restaurant performance.

Key elements:

- Top 10 Locations by Average Rating
- Average Rating by Online Ordering
- Average Rating by Table Booking
- Top 5 Restaurant Types by Average Rating

- Location Filter

Purpose: To compare customer ratings across locations, restaurant types, and service availability.

Page 3: Market Opportunities

This page focuses on dining costs, price segments, and opportunities.

Key elements:

- Restaurant Distribution by Price Segment
- Top 10 Locations by Average Dining Cost
- Location-wise Rating and Cost Comparison

Price categories used:

- **Budget:** Below ₹500
- **Mid-range:** ₹500–₹1,000
- **Premium:** Above ₹1,000
- **Unknown:** Missing cost information

Purpose: To explore affordability and compare restaurant ratings with dining costs.

Key Business Insights & Business Recommendations

8. Key Business Insights

Insight 1: Budget Restaurants Dominate the Market

The analysis shows that **54.73%** of restaurant records belong to the budget segment (below ₹500 for two). Mid-range restaurants (₹500–₹1,000) account for **34.39%**, while premium restaurants (above ₹1,000) represent **10.21%**. The remaining **0.67%** have unknown dining costs.

Business Insight: The market dataset is heavily concentrated in affordable dining options. Restaurants operating in the budget segment may face significant competition and need to differentiate themselves through food quality, service, and value for money.

Insight 2: Online Ordering Is Widely Available

Approximately **59%** of restaurant records offer online ordering.

Business Insight: Online ordering is an important service feature in Bangalore's restaurant market. Businesses should consider digital ordering capabilities when evaluating how to reach customers.

Insight 3: Table Booking Is Less Common

Only **12.47%** of restaurant records offer table booking services.

Business Insight: Table booking is considerably less common than online ordering. Its relevance may depend on the restaurant's format, dining experience, and target customers.

Insight 4: Average Restaurant Rating and Dining Cost

The overall average customer rating is **3.70**, while the average dining cost is approximately **₹555.43 for two people**.

Business Insight: These values provide useful benchmarks for comparing restaurant pricing and customer ratings across different locations.

Insight 5: Sankey Road Has High Average Dining Costs

Sankey Road has an average dining cost of approximately **₹2,505.56 for two people**, making it one of the most expensive locations shown in the dashboard.

Business Insight: Dining costs vary considerably by location. Areas with higher average costs may be worth investigating for premium restaurant concepts, although higher prices do not automatically indicate higher demand or profitability.

Insight 6: Lavelle Road Shows Strong Average Ratings

Lavelle Road has an average customer rating of **4.14**, with an average dining cost of approximately **₹1,307.93 for two people**.

Business Insight: Lavelle Road demonstrates relatively strong average customer ratings alongside above-average dining costs. This may be useful when examining the relationship between restaurant pricing and customer experience.

Insight 7: Ratings and Costs Vary Across Locations

The location-wise comparison shows differences in average ratings and dining costs across Bangalore.

For example, Koramangala 3rd Block has an average rating of **4.02** and an average dining cost of approximately **₹778.47 for two people**.

Business Insight: Some locations combine relatively strong ratings with more moderate dining costs. These locations may be useful starting points for investigating value-for-money restaurant opportunities.

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9. Business Recommendations

Recommendation 1: Develop a Competitive Budget Restaurant Strategy

Since 54.73% of restaurant records belong to the budget segment, businesses targeting affordable dining should focus on maintaining food quality, efficient operations, and competitive pricing.

Rather than competing only on low prices, restaurants should aim to provide consistent value and a positive customer experience.

Recommendation 2: Evaluate Online Ordering Opportunities

With approximately 59% of restaurant records offering online ordering, restaurant owners should evaluate whether digital ordering could improve customer convenience and market reach.

Before adopting this service, businesses should consider delivery costs, platform commissions, operational capacity, and customer demand.

Recommendation 3: Use Location-Based Analysis Before Expansion

Restaurant owners should compare location-wise dining costs, customer ratings, and restaurant concentration before opening a new outlet.

Locations such as Sankey Road may deserve further research for premium dining, while areas with strong ratings and moderate costs may be worth investigating for value-focused concepts.

Recommendation 4: Improve Customer Experience and Ratings

The overall average rating of 3.70 can serve as a reference point for evaluating restaurant performance.

Businesses should regularly analyze customer feedback and prioritize improvements in food quality, hygiene, service speed, and overall dining experience.

Recommendation 5: Select Table Booking Services Based on Restaurant Type

Because only 12.47% of restaurant records offer table booking, businesses should not assume that reservation services are equally important for every restaurant.

Fine-dining and dine-in-focused restaurants may benefit from assessing reservation demand, while takeaway-oriented businesses may prioritize other services.

Recommendation 6: Explore Value-for-Money Market Opportunities

Locations such as Koramangala 3rd Block, which shows an average rating of 4.02 and an average dining cost of approximately ₹778.47, can be shortlisted for further analysis.

However, businesses should validate these opportunities using additional information such as customer footfall, restaurant competition, rental costs, and local demand.

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10. Project Limitations

- The dataset may contain multiple records for the same restaurant.
- Some customer ratings and dining costs are missing.
- Average ratings may be less reliable for locations with very few rated restaurants.
- The dataset does not include actual revenue, profit, order volumes, or operating expenses.
- The findings describe the available dataset and should not be treated as a complete picture of Bangalore's current restaurant market.

11. Skills Demonstrated

Through this project, I practiced:

- Data cleaning and preparation using Python
- Handling missing values
- SQL business queries and aggregation
- Creating calculated columns and DAX measures

- Building interactive Power BI dashboards
- Data visualization and storytelling
- Translating analytical results into business recommendations

12. Conclusion

The Zomato Bangalore Market Analysis highlights a restaurant market dataset dominated by budget and mid-range dining options, with online ordering more widely available than table booking.

The analysis also shows that customer ratings and dining costs vary across locations. These differences provide useful starting points for understanding competition, evaluating pricing strategies, and identifying potential restaurant market opportunities.

Overall, the project demonstrates how Python, SQL, and Power BI can transform restaurant data into actionable business insights that support more informed decision-making.